Elementary row operations

A basic technique to solve any given system of equations is to convert it into a simpler system of equations by the **method of elimination**. The simpler system of equations we are looking for is a **triangular system** of equations.

However, our efforts in doing so is valid only if the solutions remain unchanged, that is, any solution of the simpler system is also the solution of the given system of equations.

The following three operations, called elementary row operations are used to simplify a given system of equations without any change in the solutions

1. Interchange any two rows
2. Multiply any row by a non-zero constant
3. Add a constant multiple of a row to another equation

Gauss Elimination Method

Example 1. Solve the following system by Gaussian Elimination method

$$\begin{aligned} x + y + z &= 4 \\ 2x + 2y + 5z &= 11 \\ 4x + 6y + 8z &= 24 \end{aligned}$$

We first form the augmented matrix, the 3×4 matrix

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 2 & 5 & 11 \\ 4 & 6 & 8 & 24 \end{array}\right)$$

Our goal in Gaussian elimination method is to use the elementary row operations to convert the part of the augmented matrix consisting of the coefficients

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & * \\ 2 & 2 & 5 & * \\ 4 & 6 & 8 & * \end{array}\right)$$

into a triangular form.

Note 1. A note on notations.

- The notation $R_p : R_p + sR_q$ implies that we replace the pth row with the sum of pth row and s times the qth row.
- The notation $R_p \leftrightarrow R_q$ implies that the pth row and the qth row are interchanged.

The method of Gaussian elimination is as follows.

$$\begin{pmatrix} 1 & 1 & 1 & 4 \\ 2 & 2 & 5 & 11 \\ 4 & 6 & 8 & 24 \end{pmatrix} \xrightarrow[R_3 : R_3 - 4R_1]{R_2 : R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 1 & 4 \\ 0 & 0 & 3 & 3 \\ 0 & 2 & 4 & 8 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & 4 \\ 0 & 2 & 4 & 8 \\ 0 & 0 & 3 & 3 \end{pmatrix}$$

The basic idea is to make all the entries under the $(1,1)$ entry equal to zero. Then make all the entries under the $(2,2)$ entry equal to zero and so on. The final matrix corresponds to the equations $3z = 3$, $2y + 4z = 8$ and $x + y + z = 4$ which we solve by backward substitution:

$$\begin{aligned} 3z = 3 &\Rightarrow z = 1; \\ 2y + 4z = 8 &\Rightarrow 2y = 8 - 4 \Rightarrow y = 2; \\ x + y + z = 4 &\Rightarrow x = 4 - 2 - 1 \Rightarrow x = 1 \end{aligned}$$

Hence the solution is the 3-tuple $(x, y, z) = (1, 2, 1)$ and the solution set is the singleton set $S = \{(1, 2, 1)\}$.

Example 2. Solve the following system of equation by Gauss elimination method.

$$\begin{aligned} x + y - z &= 1 \\ 2x - y + z &= -1 \\ -x - y + 3z &= 2 \end{aligned}$$

Form the augmented matrix and perform the row operations.

$$\left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & -1 & 1 & -1 \\ -1 & -1 & 3 & 2 \end{array}\right) \xrightarrow[R_2:-2R_1+R_2]{R_3:R_1+R_3} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & -3 & 3 & -3 \\ 0 & 0 & 2 & 3 \end{array}\right)$$

The last matrix corresponds to the system

$$2z = 3, \qquad -3y + 3z = -3, \qquad x + y - z = 1$$

The solution is $z = 3/2$, $-3y = -3 - 9/2 \Rightarrow y = 5/2$, $x = 1 - 5/2 + 3/2 = 0$. Hence the solution is $(0, 5/2, 3/2)$ and the solution set is $S = \{(0, 5/2, 3/2)\}$.

As the next example shows, in some cases we might not be able to get a *perfect* triangular form. In such cases, we have an **infinite number of solutions**.

Example 3. Solve the following system of equations.

$$\begin{aligned} 2x + 3y + 8z &= 6 \\ 2x + y + 2z &= 4 \\ 6x - y - 6z &= 8 \end{aligned}$$

Form the augmented matrix and perform the row operations.

$$\left(\begin{array}{ccc|c} 2 & 3 & 8 & 6 \\ 2 & 1 & 2 & 4 \\ 6 & -1 & -6 & 8 \end{array}\right) \xrightarrow[R_2:R_1-R_2]{R_3:R_3-3R_1} \left(\begin{array}{ccc|c} 2 & 3 & 8 & 6 \\ 0 & 2 & 6 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The corresponding system of equation is

$$2y + 6z = 2, \qquad 2x + 3y + 8z = 6$$

In such cases, we have to introduce free variables. Let $z = s$ where s is a real number. Then $y = 1 - 3s$ and $x = (6 - 8s - 3(1 - 3s))/2 = (3 + s)/2$. We have described x, y, z in terms of a single free variable s and the solution set is

$$S = \left\{ \left(\frac{3+s}{2}, 1 - 3s, s \right) | s \in \mathbb{R} \right\} = \left\{ \left(\frac{3+s}{2}, 1 - 3s, s \right) | s \in \mathbb{R} \right\}$$
$$= \left\{ \left(\frac{3}{2}, 1 - 3s, 0 \right) + \left(\frac{s}{2}, -3s, s \right) | s \in \mathbb{R} \right\} = \left\{ \left(\frac{3}{2}, 1 - 3s, 0 \right) + s \left(\frac{1}{2}, -3, 1 \right) | s \in \mathbb{R} \right\}$$

Since s is a real number, there is a solution of the given system of equation corresponding to each value of s and hence an infinite number of solutions.

Example 4. This example will show that in some cases we require more than one free variable to describe the solution. Solve the following system of equation.

$$\begin{aligned} x - y + 5z + w &= 0 \\ 2x - 3y + z - w &= 0 \\ 3x - 4y + 6z &= 0 \end{aligned}$$

Form the augmented matrix and perform row operations.

$$\left(\begin{array}{cccc|c} 1 & -1 & 5 & 1 & 0 \\ 2 & -3 & 1 & -1 & 0 \\ 3 & -4 & 6 & 0 & 0 \end{array} \right) \xrightarrow[R_2 : R_2 - 2R_1]{R_3 : R_3 - 3R_1} \left(\begin{array}{cccc|c} 1 & -1 & 5 & 1 & 0 \\ 0 & -1 & -9 & -3 & 0 \\ 0 & -1 & -9 & -3 & 0 \end{array} \right) \xrightarrow{R_3 : R_3 - R_1} \left(\begin{array}{cccc|c} 1 & -1 & 5 & 1 & 0 \\ 0 & -1 & -9 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

The corresponding equations are

$$-y - 9z - 3w = 0, \qquad x - y + 5z + w = 0$$

To solve, we let $z = s$ and $w = t$ be the two free variables. Then $y = -9s - 3t$ and $x = -9s - 3t - 5s - t = -14s - 4t$. The solution set is

$$S = \{(-14s - 4t, -9s - 3t, s, t) | s, t \in \mathbb{R}\} = \{(-14s, -9s, s, 0) + (-4t, -3t, 0, t) | s, t \in \mathbb{R}\}$$
$$= \{s(-14, -9, 1, 0) + t(-4, -3, 0, 1) | s, t \in \mathbb{R}\}$$

Again, we have an infinite number of solutions.

The last variety is the case when the solution does not exist, that is, the system of equation is inconsistent.

Example 5. Solve the following system of equation

$$\begin{aligned} x - y + 3z &= 3 \\ -2x + 2y - 6z &= 6 \\ y - 5z &= -3 \end{aligned}$$

Row operations give

$$\left(\begin{array}{ccc|c} 1 & -1 & 3 & 3 \\ -2 & 2 & -6 & 6 \\ 0 & 1 & -5 & -3 \end{array} \right) \xrightarrow{R_2 : R_2 + 2R_1} \left(\begin{array}{ccc|c} 1 & -1 & 3 & 3 \\ 0 & 0 & 0 & 12 \\ 0 & 1 & -5 & -3 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & -1 & 3 & 3 \\ 0 & 1 & -5 & -3 \\ 0 & 0 & 0 & 12 \end{array} \right)$$

Note that the last row corresponds to the equation $0x + 0y + 0z = 12$ which is not possible for any real values of x, y, z. Hence a solution does not exist.

Gauss Jordan Method

The Gauss Jordan method is a refinement of Gauss elimination method, in which we do the following.

- Make the first non-leading entry of each row equal to 1.
- Make all entries in the column containing the first non-zero entry equal to zero.

Example 6. Find the solution set for the following system of linear equations

$$\begin{aligned} x_1 + 3x_2 - 2x_3 &= 3 \\ 2x_1 + 6x_2 - 2x_3 + 4x_4 &= 18 \\ x_2 + x_3 + 3x_4 &= 10 \end{aligned}$$

The method is to proceed with the row operations to first obtain the Gauss elimination form of the augmented matrix and then do the refinements.

$$\begin{pmatrix} 1 & 3 & -2 & 0 & 3 \\ 2 & 6 & -2 & 4 & 18 \\ 0 & 1 & 1 & 3 & 10 \end{pmatrix} \xrightarrow{R_2:R_2-2R_1} \begin{pmatrix} 1 & 3 & -2 & 0 & 3 \\ 0 & 0 & 2 & 4 & 12 \\ 0 & 1 & 1 & 3 & 10 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 3 & -2 & 0 & 3 \\ 0 & 1 & 1 & 3 & 10 \\ 0 & 0 & 2 & 4 & 12 \end{pmatrix}$$

$$\xrightarrow{R_3:(1/2)R_3} \begin{pmatrix} 1 & 3 & -2 & 0 & 3 \\ 0 & 1 & 1 & 3 & 10 \\ 0 & 0 & 1 & 2 & 6 \end{pmatrix}$$

Observe that the first non-zero entry in each row is now equal to 1. Our next set of row operations would be to make the entries in the columns containing these 1's equal to zero.

$$\begin{pmatrix} 1 & 3 & -2 & 0 & 3 \\ 0 & 1 & 1 & 3 & 10 \\ 0 & 0 & 1 & 2 & 6 \end{pmatrix} \xrightarrow{R_1:R_1-3R_2} \begin{pmatrix} 1 & 0 & -5 & -9 & -27 \\ 0 & 1 & 1 & 3 & 10 \\ 0 & 0 & 1 & 2 & 6 \end{pmatrix} \xrightarrow[R_2:R_2-R_1]{R_1:R_1+5R_3} \begin{pmatrix} 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & 1 & 4 \\ 0 & 0 & 1 & 2 & 6 \end{pmatrix}$$

The final augmented matrix corresponds to the system

$$x_1 + x_4 = 3; \quad x_2 + x_4 = 4; \quad x_3 + 2x_4 = 6$$

Let $x_4 = t$, t a real number. Then $x_3 = 6 - 2t$, $x_2 = 4 - t$, $x_1 = 3 - t$ and the solution set is

$$\begin{aligned} S &= \{(x_1, x_2, x_3, x_4) = (3 - t, 4 - t, 6 - 2t, t) : t \in \mathbb{R}\} \\ &= \{(x_1, x_2, x_3, x_4) = (3, 4, 6, 0) + t(-1, -1, -2, 1) : t \in \mathbb{R}\} \end{aligned}$$

Hence an infinite number of solutions.

Example 7. Solve the following system of equations

$$\begin{aligned} x - y + z &= 3 \\ x + 2y + 2z &= 6 \\ x + 5y + 3z &= 1 \end{aligned}$$

We start with row operations on the augmented matrix.

$$\begin{pmatrix} 1 & -1 & 1 & 3 \\ 1 & 2 & 2 & 6 \\ 1 & 5 & 3 & 1 \end{pmatrix} \xrightarrow[R_3:R_3-R_1]{R_2:R_2-R_1} \begin{pmatrix} 1 & -1 & 1 & 3 \\ 0 & 3 & 1 & 3 \\ 0 & 6 & 2 & -2 \end{pmatrix} \xrightarrow[R_3:(1/6)R_3]{R_2:(1/3)R_2} \begin{pmatrix} 1 & -1 & 1 & 3 \\ 0 & 1 & 1/3 & 1 \\ 0 & 1 & 1/3 & -1/3 \end{pmatrix}$$

$$\xrightarrow{R_3:R_3-R_2} \begin{pmatrix} 1 & -1 & 1 & 3 \\ 0 & 1 & 1/3 & 3 \\ 0 & 0 & 0 & -4/3 \end{pmatrix}$$

Observe that the last row of the final augmented matrix corresponds to $0x + 0y + 0z = -4/3$ which is not possible for any x, y, z. Hence the given system of equation has no solution.

Note 2. Let us consolidate.

- If we obtain a perfect triangular form of coefficients in the augmented matrix, a unique solution exists.
- If not, we need to describe the solution in terms of free variables and hence an infinite number of solutions exist. *This always happens if the number of equations is less than the number of variables.*
- If a row of the form $(0, 0, \cdots, 0|b)$, $b \neq 0$, appears in the augmented matrix after a sequence of row operations, the given system does not have a solution.